

Course Name: Computer Architecture II

Instructor: Dr.Samer Arandi

Student: Omar Abumazen Reg Number: 12112539

Part 1: Hardware and Software Details

Name	Type
CPU	11th Gen Intel(R) Core(TM) i7-1165G7 @ 2.80GHz (2.80 GHz)
RAM Size	16.0 GB (15.7 GB usable)
L1 Cache	320 KB
L2 Cache	5 MB
L3 Cache	12 MB
O.S	WSL – Windows Subsystem for Linux.

Part 2: Table of Data .

	Vector-Vector Multiplication		
Input-size	Scalar	Vector	improvement(%)
128	0.000001	0.000000	0
256	0.000001	0.000000	0
512	0.000002	0.000001	200%
	Matrix-Vector Multiplication		
Input-size	Scalar	Vector	improvement(%)
128	0.000123	0.000062	198.38%
256	0.000138	0.000062	222.58 %
512	0.000522	0.000243	214.81 %
	Matrix-Matrix Multiplication		
Input-size	Scalar	Vector	improvement(%)
128	0.004765	0.002685	177.46 %
256	0.034878	0.021014	165.97 %
512	0.287322	0.165271	173.84 %
Improvement(%) = Scalar/Vector			
When compiling use: gcc -O0			

